Mountain Waves in the Sky: Overshooting Top Dynamics above Supercell Thunderstorms

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Final Report

Abstract

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Acknowledgements

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1 Introduction

Stratospheric water vapour (SWV) is an important climate feedback, with the ability to potentially impact surface warming [1], hamper ozone recovery [2] and decrease the intensity of the stratospheric circulation [3], influencing the global ozone distribution. Thus, it is critical to understand how the SWV budget in order to quantify these feedbacks. Unfortunately, global climate models currently struggle with this, with an average 0.5 ppmv underestimation of ppmv at 70hPa alongside a large model spread [4]. An important secondary source of SWV is mid-latitude convection from supercell thunderstorms. These storms produce overshooting tops (OTs), deep convection which overshoots the level of neutral buoyancy and penetrates up into the stratosphere. With strong storm-relative prevailing winds in the stratosphere large amounts of water vapour within this OT can be injected irreversibly into the stratosphere in the form of a turbulent hydraulic jump [5]. However, this hydraulic jump is a relatively new discovery and, as of yet, very few studies have taken to analyzing it.

1.1 AACP Overview

The SWV feedback on surface temperature has been a source of contention in recent times. In 2001, [2] found that the combined water vapour - ozone feedback due to changes from 1959 to 1999 led to a net 0.07 Kelvin surface warming per decade. This was despite the ozone feedback on its own having a net cooling effect. In 2013, [1] demonstrated that increases in SWV due to increases in tropospheric temperature causes a $0.3W/m^2$ increase in surface flux per Kelvin of surface warming. A third of this was found to be due to water vapour entering the stratosphere through the tropical tropopause layer, with the remaining two thirds being due to water vapour entering through extra-tropics. However, recent studies challenge this claim. For example, using a multi-model analysis, He at al. (2025) [6] claimed that after both stratospheric cooling and tropospheric warming feedbacks associated with SWV changes are included, the total radiative forcing due to SWV is negligible. The uncertainty around the radiative feedback of SWV further increases the importance of accurately calculating its budget, and finding how it will change under climate change, for future studies.

Water vapour can travel up into the stratosphere in many different ways. The main method of transport is through the tropical cold point tropopause [4], although transport from volcanic eruptions [7] and convective overshooting are important secondary sources.

Focusing on convective overshooting, Dauhut et al. (2018) [8] used a model to show that tropical convective storms, such as the "Hector the Convector" can inject over 2700 tons of water vapour into the stratosphere over the course of an hour, with this injection being poorly resolved by convective parameterizations. In the midlatitudes, an aircraft flown 20 hours after a supercell thunderstorm showed that large 1-2 ppmv enhancements in water vapour were found well above the tropopause (19.25 km up!) [9]. Additionally, model simulations have shown that water vapour injected into the stratospheric overworld (above the 380K potential temperature isoline) by OTs tends to stay even after the supercell system which triggered them is gone [10]. This injection is not well resolved by current generation satellites such as Aura's Microwave Limb Sounder.

In about 10% of cases, storms produce an above anvil cirrus plume (AACP), with AACP's being formed by 75% of supercells. Despite this, AACPs produce 57% of all severe weather reports and 73% of significant severe weather reports [11]. AACP's are characterized by a unique texture distinct from the storm larger anvil several kilometers underneath, as well as as a high IR brightness temperature (but not always)! AACPs precede severe weather reports by 31 minutes on average, and this one feature predates U.S National Weather Service warnings 25% of the time. It has also been shown that AACP storms inject more water into the stratospheric overworld [5] [12] [13], although whether they inject more into the stratospheric middleworld is disputed. High storm-relative winds in the lower stratosphere and increasing low-level instability have been found to be environmental conditions which favour AACP formation [13].

The mechanism behind AACP formation is the breaking of gravity waves forced by the OT [5] [8] [14] [15] [16]. The OT acts similarly to a mountain at the surface, blocking the incoming stratiform flow, and forcing the winds up and over the OT. The flow then accelerates over this "mountain", continuing to accelerate as it travels over the leeside of the OT. These waves break, and the accelerated flow reconnects to the ambient downstream flow in the form of a turbulent hydraulic jump [5]. This hydraulic jump takes the high levels of water perturbation from the OT (which originated in the upper troposphere [15]) and mixes it into the stratosphere (see O'Neill et al. (2021) [5] figure 2B). The cloud ice "jumped" from the OT is what forms the AACP. In subsaturated environments, this cloud ice will then sublimate, irreversibly adding water vapour into the stratosphere. However, in supersaturated environments the vapour will actually deposit on the ice crystals causing the stratosphere to lose moisture when the ice precipitates out [15]. When the storm ends, a lot of this water vapour which hasn't been mixed in this fashion will fall back into the troposphere with the OT [10]. In their 1km grid spacing simulation, Qu et al. (2020) [16] found that wave breaking accounted for 59% of the vertical vapour injection via vertical transport while the mean updraft accounted for 39%. However, this ratio is expected to skew further towards wave breaking as the simulation resolution is increased and more wave breaking is resolved.

1.2 Mountain Waves

With the overshooting top acting as a blockage to oncoming stratiform flow, it stands to reason that theory describing fluid flow over other topography can be used. This has been studied since at least the 1950s [17], with a significant bulk of research completed on the topic since then.

A lot of this research has involved shallow water flow, and while this is a simpler case then the stratified flow seen in the atmosphere, it can still be used to better understand the theory. With this, the steady state shallow water equations can be used to describe the flow [18]:

$$u\frac{\partial u}{\partial x} + g\frac{\partial D}{\partial x} + g\frac{\partial h}{\partial x} = 0, \tag{1}$$

$$\frac{\partial uD}{\partial x} = 0. \tag{2}$$

Here, D is the fluid depth and h is the obstacle height. Equation 1 describes momentum continuity, where momentum via horizontal advection (the first term) of the fluid is balanced by gravitational potential from either the fluid piling up with increased fluid depth (second term) or as the fluid rising as it is forced over the obstacle (third term). Equation 2 describes mass continuity where changing fluid velocity over x must be accompanied by a corresponding change in the fluid depth. Manipulating we see that [19]:

$$(1 - \mathrm{Fr}^2)\frac{\partial D}{\partial x} = -\frac{\partial h}{\partial x}, \tag{3}$$

where Fr is the Froude number, and in this case is U/gD. From equation 3, we can see that if the upstream Froude number is less than one (middle of Fig 1), mountain height increases the fluid depth must decrease. As fluid depth decreases, the fluid speed increases over the mountain by mass continuity. Similarly, for an upstream Froude number over one (top of Fig 1), the fluid acts like a ball going over a hill, slowing down and thickening as it goes over the mountain before accelerating down the other side. However, in a certain regime of subcritical upstream Froude number, with a tall enough mountain (see Fig 2.9 of Baines (2022) [20]), the fluid accelerates enough to let the Froude number transition from subcritical to supercritical over the mountain (bottom of Fig 1). This causes the fluid

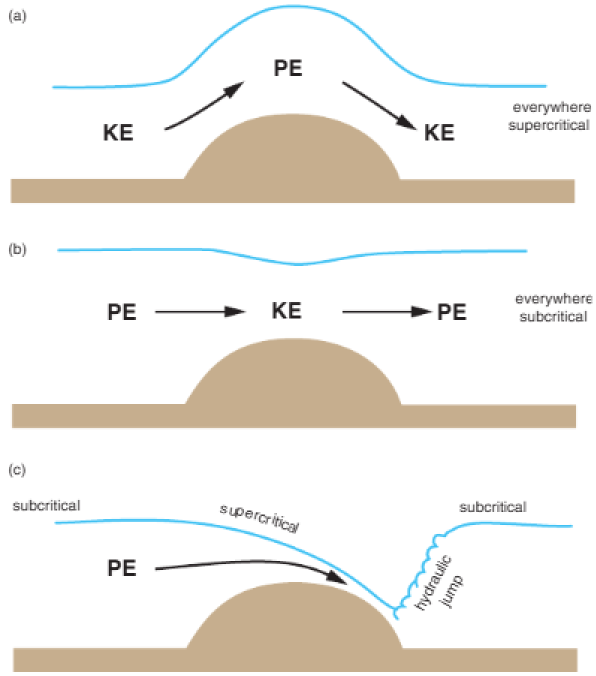

Figure 1: Figure 12.12 from Markowski and Richardson (2011) [19]. Flow over an obstacle for a single layer fluid. Top panel is for supercritical flow everywhere (Fr ¿ 1), the middle panel is for subcritical flow everywhere (Fr ¡ 1), and the bottom panel shows a transition from subcritical to supercritical over the barrier.

accelerating all the way over and down the mountain, creating a downslope windstorm. The fluid then recovers to the ambient flow in the form of a turbulent hydraulic jump. However, this Froude number definition is suspect at best for an atmospheric context [19] as the depth of the layer is not well defined for stratified flows.

In a simulation of two layer flow with a transition layer past topography, Rotunno and Bryan (2018) [21] [22] show that steepening of leeside isentropes baroclinically produces vorticity. This baroclinically produced vorticity is transported further downstream where it causes the isentropes to overturn. This isentropic overturning causes a location of local instability, the wave breaks, and the transition layer mixes in with itself forming a hydraulic jump. The results of this simple surface simulation (figure 3e of [21]) shows a remarkable resemblance to the result given in O'Neill et al (2021) ([5] fig 2B). This demonstrates that this surface mountain wave theory is indeed quite applicable to mountain waves in the sky.

A hydraulic jump can be thought of as a discontinuity between supercritical and subcritical flows. This is demonstrated using the QRS framework in Baines (2022) [20]. Here Q represents mass flux, R represents streamline energy (or the Bernoulli function) and S represents the momentum flux. A phase diagram of normalized S and normalized R (fig 2.20 of Baines (2022) [20]) can be created, with the upper edge of the drawn cusp representing subcritical flow and the lower edge being supercritical flow. The area within those edges represent periodic wave solutions which tend to zero amplitude as we move closer to the subcritical branch. A hydraulic jump is a horizontal line along the cusp of that figure, with the momentum being constant while energy is lost due to turbulence and viscosity. However, if R is decreased by an energy less than that required to travel across the gap, we get a wave-train. For supercritical Froude numbers less than 1.3, a tiny amount of dissipation converts the flow into large amplitude undular bores. For Froude numbers over 1.3 however, we need a lot of energy dissipation to cross an unsteady zone on the phase diagram, leaving a more turbulent structure as the leading wave breaks. For Froude numbers over about 1.55, we get a fully turbulent bore. The bore amplitude increases as we increase the Froude number from one, until the 1.3 value is hit and the leading wave breaks. Afterwards, the amplitude decreases with increasing Froude number. However, the jump amplitude for any hydraulic jump monotonically increases with Froude number.

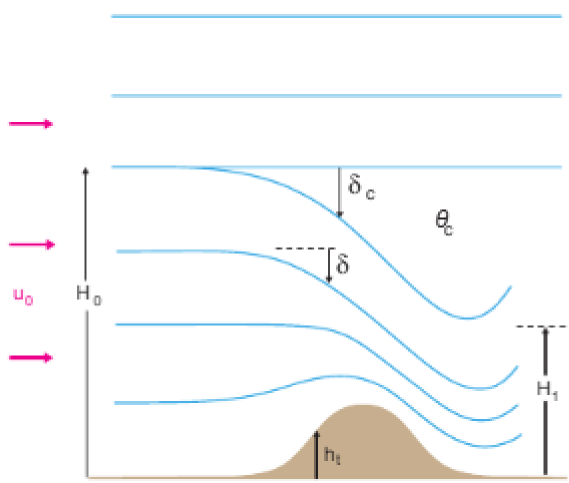

Figure 2: Figure 12.13 from Markowski and Richardson (2011) [19]. A depiction of the dividing streamline where δ_c represents the height deviation of the lower branch of the streamline from it's upstream height (H_0).

For the stratified infinite depth flows seen in the atmosphere, Smith (1985) [23] devised a theory where when standing waves overturn/break, there is a region with low stability and a critical level. A critical level occurs when the phase speed of a wave equals the wind speed, causing group velocity to become nearly horizontal and is associated with wave reflection [19]. Smith (1985) [23] specified a dividing streamline 2, which, for a tall enough mountain, descends far down towards the surface. The region below the streamline is described by steady flow, but within it is a well mixed region of constant θ. The critical layer does not allow wave energy to propagate through the mixed layer, reflecting the wave energy back down causing high speed downslope windstorms. If the dividing streamline is within a certain number of wavelengths off the ground, there is a hydraulic jump.

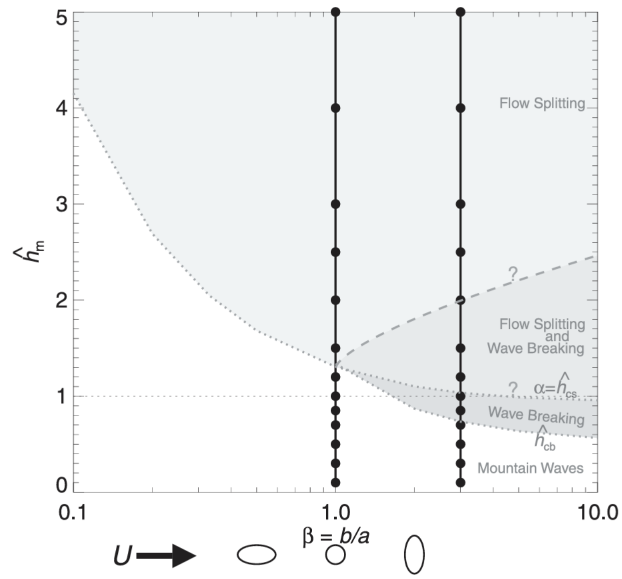

Figure 3: Figure 1 from Eckermann et al. (2010) [24]. A phase space diagram of different flow regimes for different mountain heights and shapes.

Additionally, the shape of the mountain also affects the flow type as seen in figure 3 [24]. In figure 3, $\hat{h}_m$ is the normalized mountain height equal to Nh_m/U, where U is the upstream fluid velocity, N is the Brunt–Väisälä frequency and h_m is the peak mountain height. Note that $\hat{h}_m$ is often interpreted as an inverse Froude number for these stratified flows. However, as expressed in Baines (2022) [20], there are many different Froude numbers choose from in this context. Returning to figure 3, wave breaking only occurs for values of β larger than one and for taller normalized mountain heights. However, if the mountain is too tall, the waves won't break and the flow will simply split over either side of the mountain.

1.3 Current Work

Unfortunately, despite the large amounts of observational and modeling studies on OTs and the vast array of literature pertaining to mountain waves, very few studies have done in depth dynamics modeling of OTs using mountain wave results. One of the few such studies, O'Neill et al. (2021) [5], had an extremely idealized environment for their simulation. They used updraft nudging for their initial convective initiation, alongside an idealized supercell thunderstorm sounding and a horizontally homogeneous domain. This is despite their extremely high resolution simulation, with grid spacing in all three dimensions of 50m. Nevertheless, the idealities of their simulation likely led to some artifacts, such as cloud ice suspended above the OT before the gravity wave break. This cloud ice could potentially provide a critical layer for gravity wave reflection; hence, it is imperative to understand whether it is an artifact, and if it's not, to understand what its source was. Thus, running more realistic simulations is a key goal of this work.

2 Methods

For this work, George Bryan's Cloud Model 1 (CM1) [25] was used; a 3-D non-hydrostatic numerical model designed for studying small scale atmospheric processes.

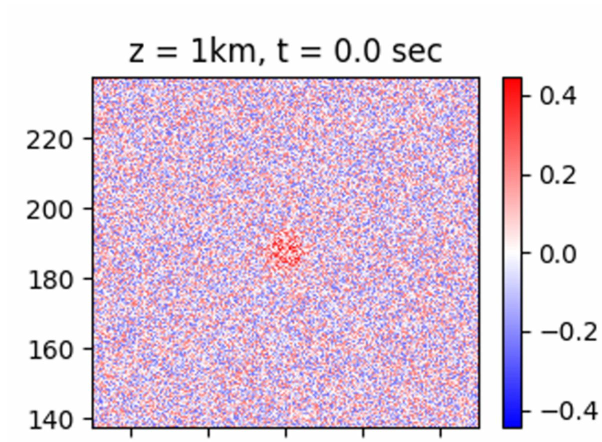

Figure 4: The potential temperature field (K) at the first timestep of our "realistic" simulations. Spatial units are in kilometers

While the model setup of O'Neill et al (2021) [5] was mostly followed, a number of modifications were made to attempt to increase the realism. First, similarly to Gordon and Homeyer (2022) [12] instead of updraft nudging, a thermal bubble in the center of the domain was used to proxy a more realistic form of convective initiation. This is likely more physical due to forming the updraft indirectly via a temperature perturbation, rather than directly via updraft nudging. However unlike Gordon and Homeyer (2022) [12], the maximum potential temperature perturbation of the bubble was set to 0.25K, a quarter of the default value, which is hopefully more representative of how convection is actually initiated in the atmosphere. Additionally, as true atmospheres are not horizontally homogeneous, 0.25K magnitude noise was added to the potential temperature field in the first time step, similar to Markowski (2020) [26]. This allows some amount of turbulence to develop in the system

[26]. The magnitude of the noise was set to match the magnitude of the bubble. A portion of the first timestep 1km up in these simulations can be seen in 4, with the red area in the centre being the thermal bubble.

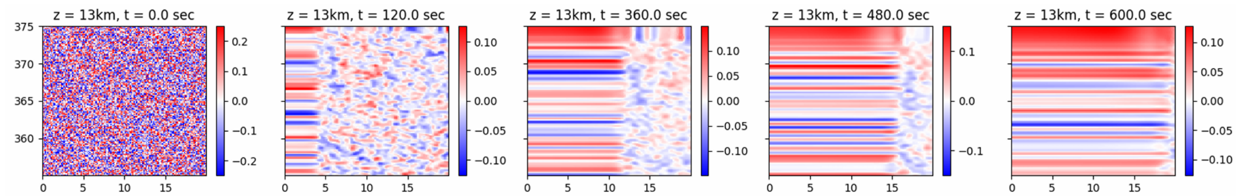

Figure 5: Similar to figure 4 but with open radiative boundary conditions and at the edge of the domain.

Unfortunately, when adding noise to the open radiative boundary conditions used by O'Neill et al (2021) and Gordon (2022) [5] [12], lines of a constant value tend to blow in from the edges as seen in figure 5. These lines protrude further into the domain over time, likely eventually interfering to some extent with the storm at the center. We attempted to remove these lines via tweaks to the simulation such as using eddy recycling. Eddy recycling involves taking a line average of a variable a specified distance from the boundary before subtracting it from the actual field to retrieve the perturbation. The perturbation is then added to the inlet profile at each time step [27]. A nice visual representation of this is given in figure 3 of Maronga et al. (2015) [27]. Unfortunately, this did little to prevent the lines as they continued to enter the domain, but this time alongside even faster moving non-line temperature perturbations which were likely an artifact of the eddy recycling. Other modifications to attempt to resolve this problem included starting with no winds before adding the hodograph afterwards (with and without eddy recycling) which likely didn't work as there is limited mixing of the noise without incoming winds. Additionally, starting with periodic boundary conditions before restarting with open boundary conditions also failed (once again with and without eddy recycling), although this time with long domain spanning lines instead of those seen in figure 5. Thus, similar to Markowski (2020) [26] periodic boundary conditions were used, but in our case with horizontal damping. This was done to prevent gravity waves which had exited from one end of the domain from re-entering through the other. However, we are unsure if Markowski (2020) [26] used periodic boundary conditions because of these lines, or if there was another reason (no justification is given in the paper).

For this work, three simulations for each of the strong and weak wind soundings in O'Neill et al/ (2021) [5], rotated to ensure that the winds were oriented parallel to the x axis in the lower stratosphere. The strong wind experiment formed a AACP in O'Neill (2021) due to the imposed high storm-relative winds, while the weak wind sounding did not, and was used as a control. Here, three middle wind simulations were also run, an average of the strong and weak wind soundings. Three simulations of each type allows us to compare the spread of possible outcomes since differing random noise has the potential to cause different results. This is exemplified in Markowski (2020) [26] where some simulations are tornadic and others aren't despite the sole difference being simulation noise.

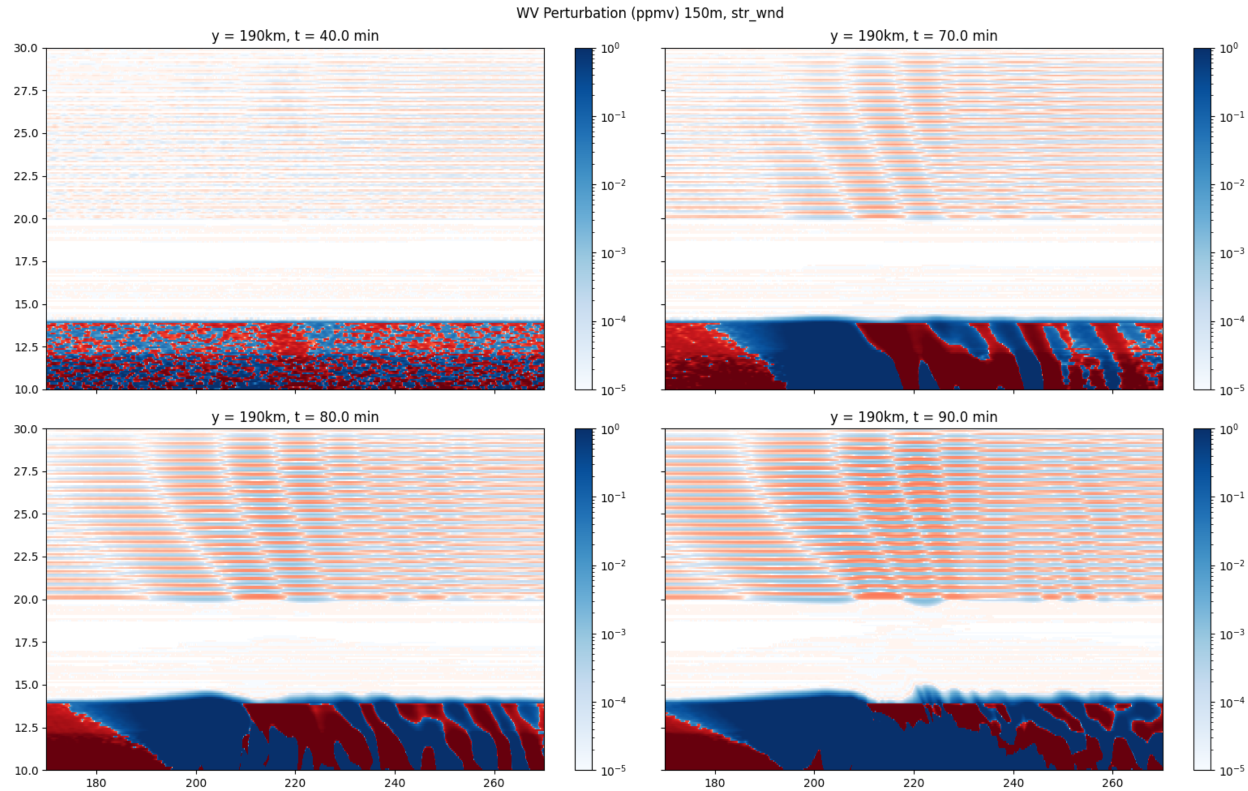

Figure 6: A water vapour perturbation plot with a saturated log scale colourbar. Blue represents positive, while red represents negative. The y-slice is at 190km in all four images.

Additionally, the soundings all have a large amount of vertical wind shear, with twisting winds. This wind shear is a key condition for supercell generation as it increases mesocyclone intensity [19]. However, upon analysis we noticed that there were lines in the water vapour field as seen in figure 6. This was due to the resolution of the sounding being 500 metres above the 20 kilometer mark, lower than that of the model. To resolve this, the sounding was interpolated using the original formulas for the profiles given in [28]. Unfortunately, O'Neill et al. (2021) [5] used these soundings and likely has this feature which does appear to grow in magnitude over time. Thankfully, the effect is likely quite small with the value of the feature at 90 minutes only appearing to go up to around a thousandth of a ppmv. The soundings are all set to have a tropopause at 12km via an inflection in the potential temperature profile. The strong and weak wind sounding profiles (wind, water vapour and potential temperature) can be found in the supplementary material of O'Neill et al. (2021) [5].

These ensemble simulations were completed with 150 metre grid spacing in the horizontal and 100m spacing in the vertical. To ensure that dynamical features were not missed by this lower resolution, an additional strong wind simulation was completed with 75 metre grid spacing in the horizontal and 50 metres in the vertical. Thus, in all simulations CM1's pressure solver number 2 was used, as it is meant for when the vertical grid spacing is approximately the same as the horizontal. This differs from O'Neill et al. (2021) [5] where they used pressure solver number 3, meant to be used from the vertical grid spacing is much less than the horizontal. Stretching of the grid cells in the vertical to better resolve boundary layer processes was considered, however doing so was found to greatly slowdown the simulation speed and thus was not implemented. For time resolution, CM1's adaptive timestep was used which changes the timestep in order to set the Courant parameter to 1. However, this was found to cause model instabilities in the centre of the storm regardless, so the timestep was made to be around 1.4 seconds at approximately the time of storm maturity.

In this simulation the turbulence model used was the CM1 default for supercells, the TKE scheme from Deardorff (1980) [29]. This scheme uses the Boussinesq approximation to model the Reynolds stresses, which assumes that viscous stresses act in a similar way to turbulent mixing. It uses a heavily parameterized turbulent kinetic energy (TKE) transport equation, which along with the length scale specifies the turbulent eddy viscosity as $\nu_t \propto k^{1/2}/\epsilon$. Here k is TKE while ϵ is the dissipation rate of TKE, specified as $\epsilon \propto k^{3/2}/\ell$, where ℓ is the turbulence length scale. Transport equations for the equivalent potential temperature and total specific humidity are also used to parameterize terms in the TKE transport equation. [30].

Turbulence above these supercell thunderstorms is invisible to pilots and thus better understanding its magnitude is critical. Pilots use a quantity called eddy dissipation rate (EDR) to quantify the turbulence in the area, given as the cube root of the TKE dissipation. Values for this parameter are typically between zero and one and tend to scale with the winds [30]. Many aircraft across North America currently calculate EDR in situ onboard their aircraft due to its importance when flying [30].

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